

Beckham Share — Build Report

A self-hosted file-sharing platform, delivered end-to-end & live in production

share.beckham.ai · FastAPI · PostgreSQL · OIDC · Docker · scope, time, cost & receipts · generated 2026-06-23

<1 day

Calendar to live

Jun 22 13:32 → Jun 23 — one sitting

5.5 h

Engineer hands-on time

34 build turns / 5 sessions

\$357

AI compute (reference*)

278M tokens · subscription

14×

Faster (engineer-hours)

vs ~80 h unassisted est.

2,388

Lines of code & tests

+ ~7,300 words of docs

28

Automated tests

24 backend + 4 E2E · CI green

The bottom line — assisted vs. an unassisted engineer

The same production scope, built two ways. “Assisted” is the actual engineer time logged on this build; “unassisted” is a transparent bottom-up estimate (broken out later). Hours are **engineer hours**.

14×

fewer engineer-hours
(5.5 h vs ~80 h)

89%

lower delivered cost
(\$912 vs ~\$8,000)

\$7,088

net saved on one project
(labor + compute)

Engineer hours to deliver the full scope

Assisted (actual)  **5.5 h**

Unassisted (est.)  **80.0 h**

Delivered cost — labor + AI compute

Assisted  **\$912**

Unassisted (est.)  **\$8,000**

Assisted cost = 5.5 engineer-hours × \$100/h + \$357 compute. Unassisted = ~80 h × \$100/h. Rate is an illustrative loaded rate; see methodology for the range.

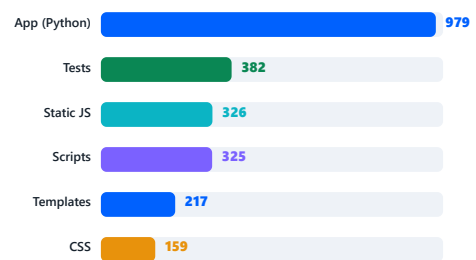
What was built

A complete, production-deployed product — not a prototype. Public anonymous sharing, an identity-gated member workspace, abuse controls, a full test pyramid, CI, and documentation.

Deliverables

- ✓ Upload → UUID share link
- ✓ Copy + email link actions
- ✓ OIDC sign-in (Authentik)
- ✓ Rate-limit + size cap
- ✓ Member-only SMTP email relay
- ✓ Docker Compose (app + db)
- ✓ 3-layer test suite + CI
- ✓ Live on 3 hostnames, valid TLS
- ✓ Configurable / editable expiry
- ✓ Public anonymous landing page
- ✓ `dropbox` -group gated workspace
- ✓ IP + browser fingerprint audit
- ✓ Brand logo + favicon
- ✓ Caddy / HAProxy wiring
- ✓ Full reference docs
- ✓ Deploy + smoke scripts

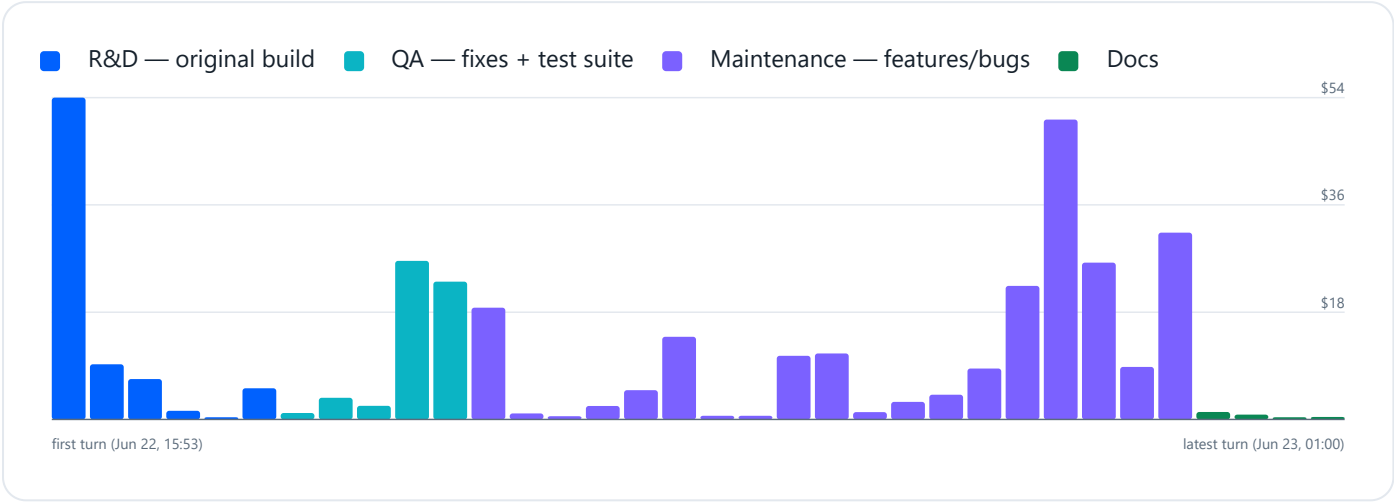
Codebase footprint



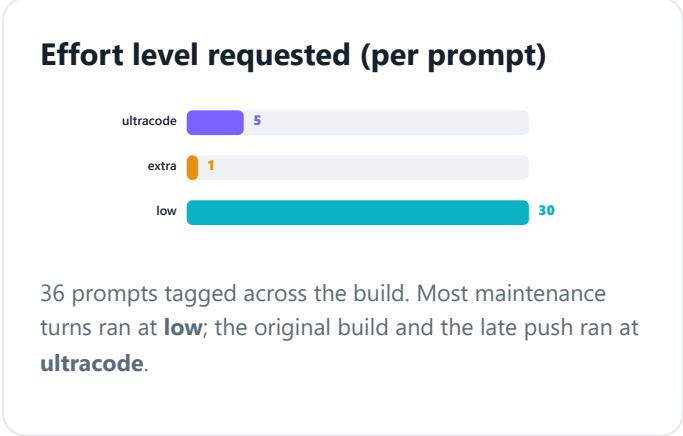
~2,388 lines of code, config & tests · 20 Python modules · 14 HTTP routes · 11 commits · ~7,300 words of documentation.

The build, by the numbers

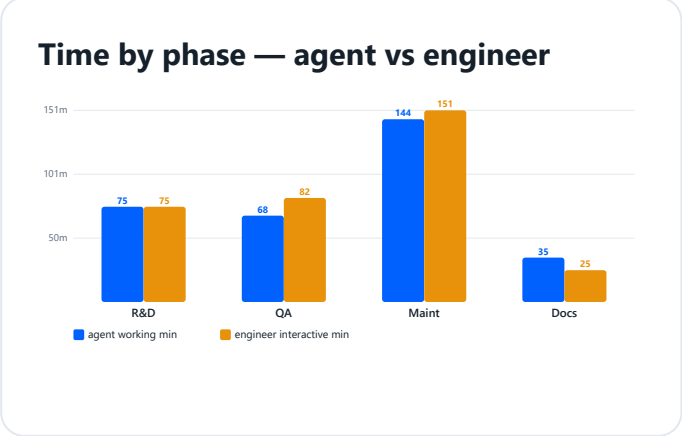
AI compute spend per turn over the ~12-hour window, coloured by phase. The tall bars are the heavy generation turns (initial build, the test-suite turn, late-night feature work).



Spend by phase			
PHASE	COMPUTE	TOKENS	SHARE
R&D	\$76.97	56.2M	22%
QA	\$56.67	52.7M	16%
Maintenance	\$220.21	167.1M	62%
Docs (so far)	\$2.71	2.1M	<1%
Total	\$357	278M	100%

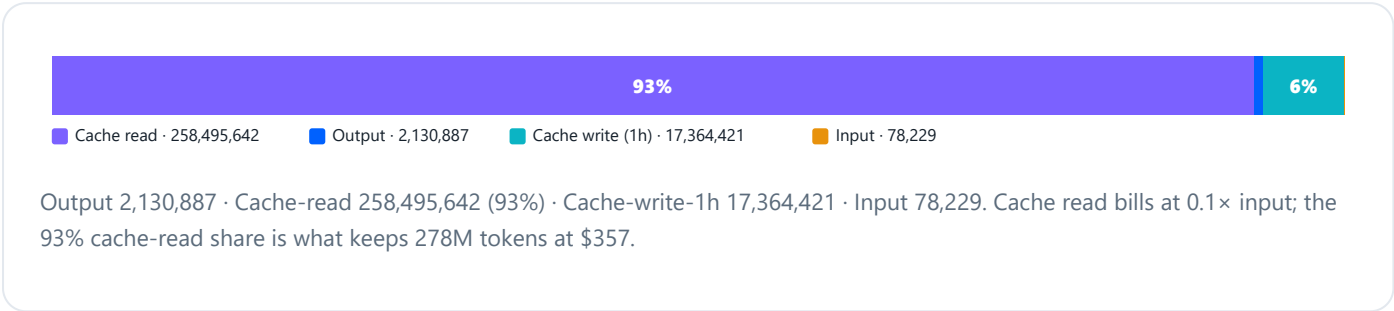


36 prompts tagged across the build. Most maintenance turns ran at **low**; the original build and the late push ran at **ultracode**.



Compute profile

Where the 278M tokens went. Agentic builds are dominated by **cache reads** — the conversation/context replayed cheaply on every tool round-trip (read at ~10% of input price), which is exactly why the effective cost stays low despite the huge token count.



Output 2,130,887 · Cache-read 258,495,642 (93%) · Cache-write-1h 17,364,421 · Input 78,229. Cache read bills at 0.1× input; the 93% cache-read share is what keeps 278M tokens at \$357.

Assisted vs. unassisted — the estimate

A bottom-up estimate of what the *same* scope would take one experienced engineer working unassisted, by work area. Ranges are shown; the comparison above uses the midpoint.

WORK AREA	LOW (H)	HIGH (H)
Backend application (models, storage, auth, fingerprint, rate-limit, email, routes)	16	24
OIDC / Authentik integration + idempotent setup script	8	12
Front-end (templates, JS, CSS, UX) + branding	10	16
Infrastructure (Docker, Caddy, HAProxy, deploy + secret scripts, live deploy)	8	12
Automated test suite (pytest + Playwright + smoke) + CI	10	16
Documentation (architecture, config, ops, security, API, contributing)	4	8
Debugging & iteration (sign-in, share modal, CI, mobile picker)	6	10
Unassisted total	62 h	98 h

Midpoint ~80 h ≈ 10.0 working days ≈ ~2 calendar weeks. Actual assisted engineer time was 5.5 h in a single day.

5.5 h

assisted engineer time

~80 h

unassisted estimate (mid)

~74 h

engineer-hours saved

Status & what's left

Done & live

- Full app deployed at **share.beckham.ai** (+ dropbox / buckets)
- OIDC sign-in working for both members; member-gated email
- Automated test suite + CI green on every push / PR
- Brand logo, favicon, Dropbox-like UX
- Reference documentation (this turn) + troubleshooting article
- Resolved: sign-in, stuck share modal, CI, mobile picker (#5)

Open / next

- enh #7** file lifecycle + download analytics
- enh #8** configurable default + editable expiry
- enh #9** folder creation / organization
- docs** open PR for the docs branch → merge
- housekeeping** commit synced proxy configs in their own repos
- polish** optional brand-purple accent

Methodology & caveats. **Compute cost** (\$357) is the stamped Actual from the local accounting ledger (`~/ .claude/time-tracking`), priced at the schedule in effect on the build date. Under a subscription login this is **API-equivalent reference value, not cash billed** — the authoritative spend is the provider console. It covers turns through the documentation phase; this report-generation turn is not yet stamped, so the live total will tick up slightly. **Engineer time** is the curated interactive time from the project ledger (15-minute away-rule applied), ~5.5 h. **Token totals** (278M) are summed over every API round-trip in all 34 turns, so cache-read dominates by design. **The unassisted baseline is an estimate**, not a measurement: a bottom-up 62–98 h range for one experienced engineer building the same scope from scratch, including deploy, tests and docs. The dollar comparison applies an illustrative loaded rate of \$100/h (a typical \$75–\$150/h range would put unassisted labor at ~\$6,000–\$12,000). Figures are receipts-for-work indicators, not an audited cost model.